

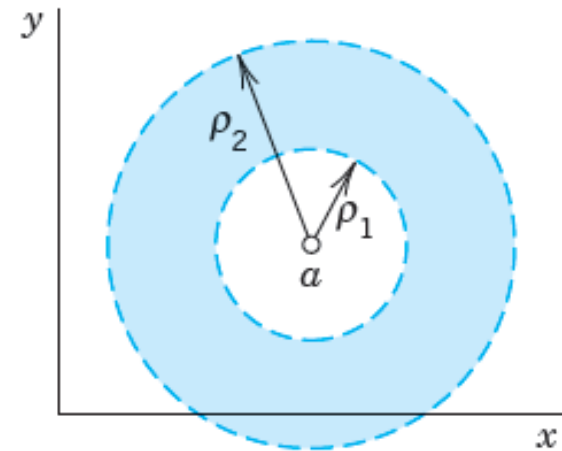
Beregningsteknik for Computer Ingeniører

MM 6: Complex functions

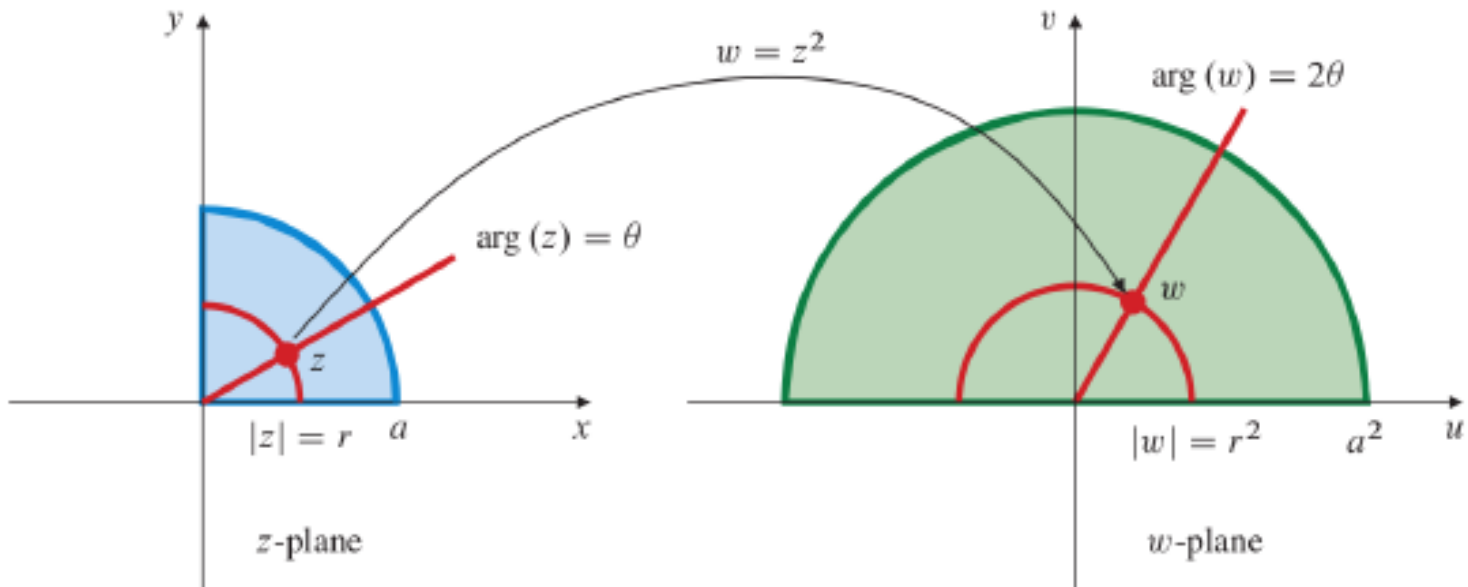
Topics: Complex functions
“Complex” calculus – derivatives and integrals

Complex functions- calculus

- A complex function $w = f(z)$ assigns to every $z \in \mathcal{C}$ a value $w \in \mathcal{C}$ where the z variable varies in the **domain** S - domain of definition - and w , the value of f at z , in the **range** of f
- An open set S has a "neighbourhood" consisting entirely of points belonging to S (because the boundary is not included, cf. closed set with boundary points included)
- **Domain:** An open and connected set S (i.e., every point in the interior can be joined by broken line segments with points in S) is called a DOMAIN;
- **Region:** the domain + possibly some boundary points

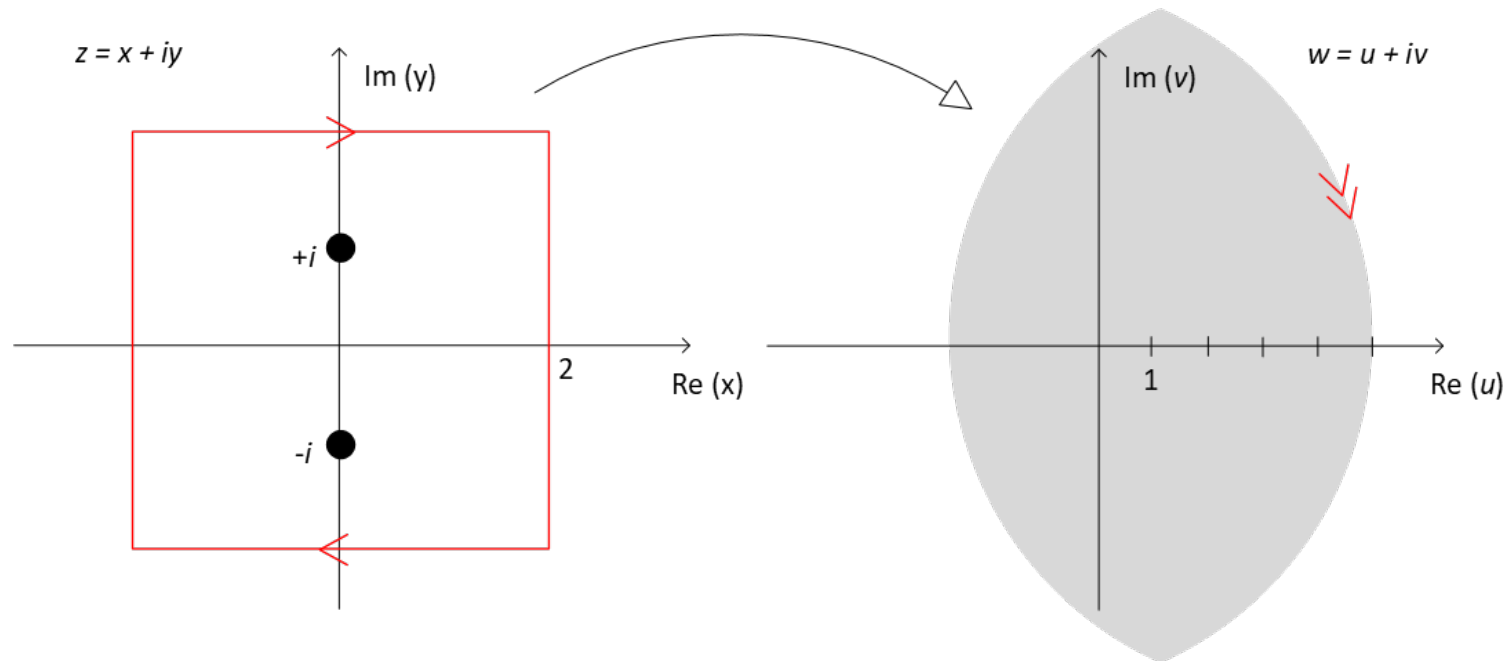


Example $f(z) = z^2$



Nyquist/Root Locus plots

- Evaluating $f(z)$ around the red path leads to twice around the gray area on the right – twice (enclosing 0) since we encircle two zeros of the function $f(z) = z^2 + 1$
- Generally, the number of encirclements equals the difference between number of zeros and poles, viz. $f(z) = z^2/(z + 1)$



Complex functions - calculus

- Can also see the complex function $w = f(z)$ as combining two real functions

$$w = f(z) = u(x, y) + iv(x, y)$$

- Similar to real-valued functions we can define the derivative at z_0

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{z - z_0}$$

- OBS Contrary to real valued functions, z may approach z_0 from ANY direction and should reach the same certain value!
- **Analyticity:** $f(z)$ is analytic in a domain D if $f(z)$ is defined and differentiable at all points in D ; f is analytic at a point $z = z_0$ if $f(z)$ is analytic in a neighbourhood of z_0 . Polynomials in z are analytic functions in D , as well as rational functions

$$f(z) = g(z)/h(z), h(z) \neq 0$$

Differentiation rules

- Differentiation is similar to real-valued functions, e.g. linearity for sums and differentiation by parts for functions multiplied and divided
- Differentiation of products (product rules):

$$(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$$

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$$

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$$

$$w = (cf(x_0))' = cf'(x_0)$$

- Differentiation of composite functions:

$$(x_0^n)' = nx_0^{n-1}$$

$$h = f \circ g; f'(g(x_0))g'(x_0)$$

Integration

- Complex integrals are called line integrals (from a direction!)

$$\int_C f(z)dz$$

- C is a smooth – continuous and with derivative different from 0 (i.e., it has a defined continuously turning tangent vector along the curve)
 - curve in the complex plane, representing the path of integration
 - can be represented by a parametrization:

$$C: z(t) = x(t) + iy(t), a \leq t \leq b$$

- Similar to the real case, the complex integral is linear and exists if $f(z)$ is continuous and C is piece-wise smooth
 - We can partition the path, reverse it (with integral sign reversed)

Integration

- Method 1 (analytic function)
 - Analytic: $f(z)$ is defined and differentiable at all points in the **simply** connected domain S

$$\int_C f(z)dz = \int_a^b f(z)dz = F(z_b) - F(z_a), F'(z) = f(z)$$

- We get the same value for all paths!

- Method 2 (continuous function)
 - C is piecewise smooth and $f(z)$ is continuous on C (*not necessarily analytic in the domain containing C*)

$$\int_C f(z)dz = \int_a^b f(z(t)) \frac{dz(t)}{dt} dt, a \leq t \leq b, z(t) \in C!$$